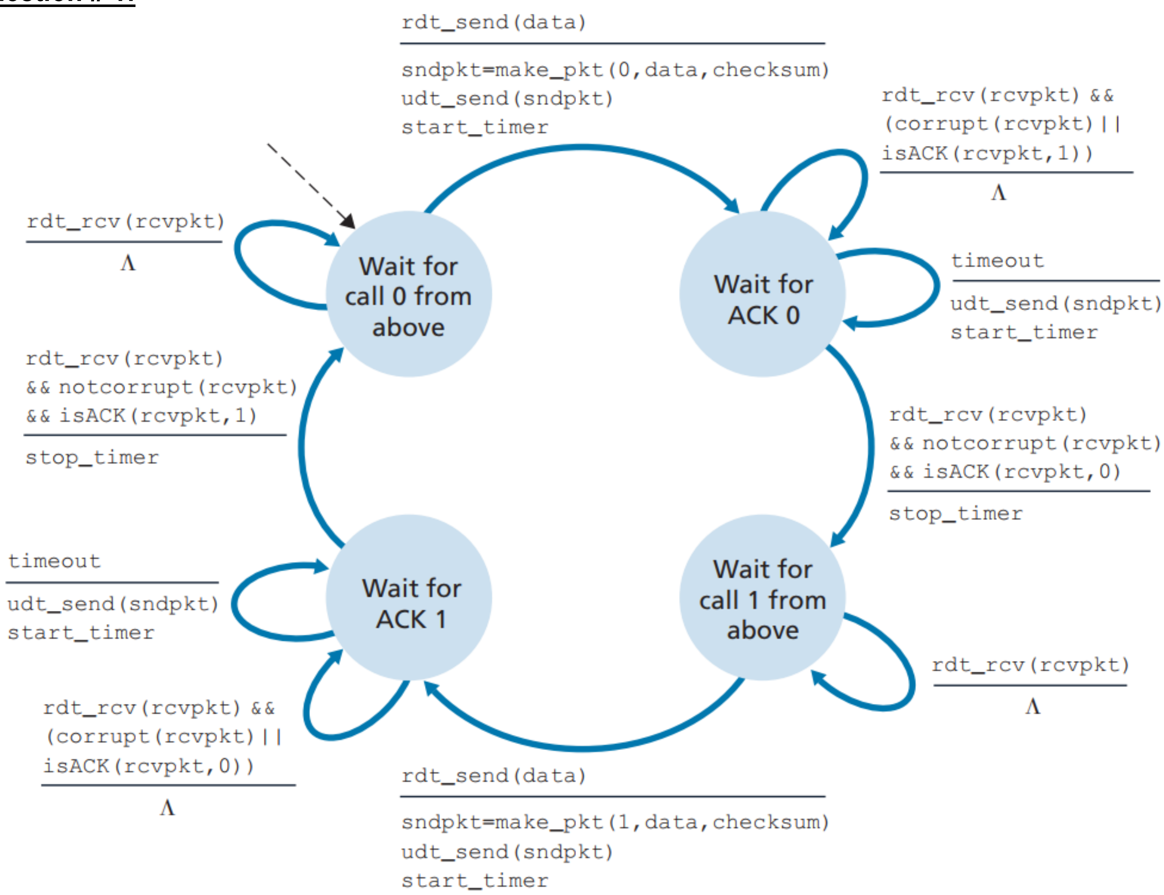
Question # 1:



Question # 2:

a)

Estimated RTT = (1 - α) * Estimated RTT + α*Sample RTT

Estimated RTT = (1 – 0.125) (350 msec) + (0.125) (380 msec) = 353.75 msec

Dev RTT = (1 - β) * Dev RTT + β * |Sample RTT – Estimated RTT|

Dev RTT = (1 – 0.25) (34 msec) + (0.25) * |380 msec – 353.75 msec| =32.06 msec ≈ 32 msec

b)

New timeout = Estimated RTT + 4 * Dev RTT = 353.75 msec + (4) (32.06 msec) = 481.99 msec ≈ 482 msec

Pervious timeout = Estimated RTT + 4 * Dev RTT = 350 msec + (4) (34 msec) = 486 msec

The new timeout value is decreased as compare to pervious timeout value.

Question # 3:

a) Routers may set bits in passing packets and may send messages to the source or destination to indicated congestion.

b) Congestion is inferred at the end hosts (sender or receiver, either as a result of packet loss or increased delay).

c) TCP uses an end-end approach

Question # 4:

a) The times where TCP is in slow start are: 1-3, 13-15, 24-26, 28-29, 31-35.
b) The times where TCP is in congestion avoidance are 4-12, 16-23,36-39.
c) The times where TCP is in fast recovery are: 18.
d) The times where TCP has a loss by timeout are: 12,23,27,30,34,39.
e) The times where TCP has a loss by triple duplicate ACK are: 17

Question # 5:

Subnet	Network Address	Custom Subnet Mask	Host Range	No. Of Hosts
Ufone	209.118.126.0/ 24	255.255.255.0	209.118.126.0 -- 209.118.126.255	254
Zong	209.118.127.0/26	255.255.255.192	209.118.127.0 --- 209.118.127.64	48
Telenor	209.118.127.128/25	255.255.255.128	209.118.127.128 --- 209.118.127.255	126